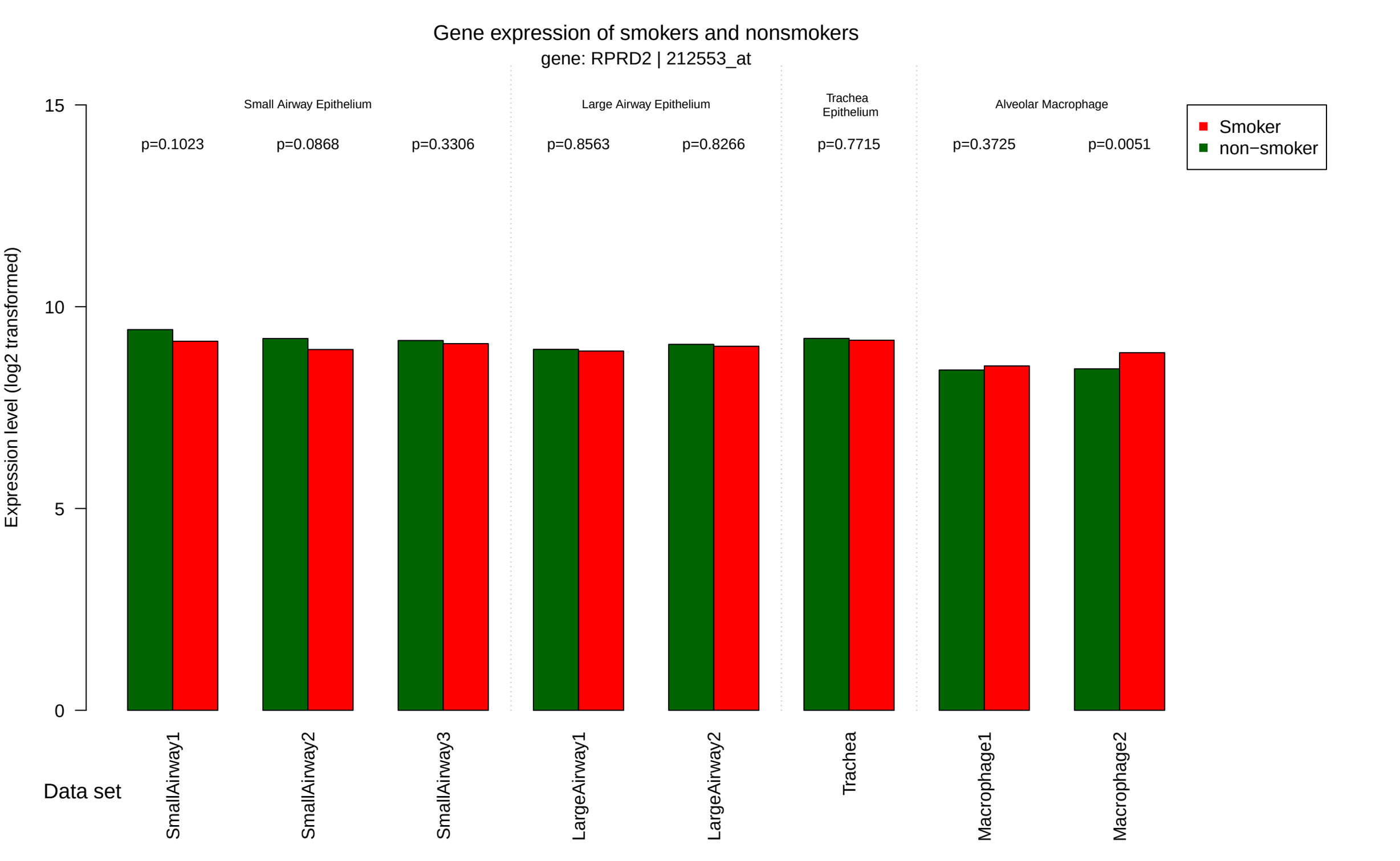
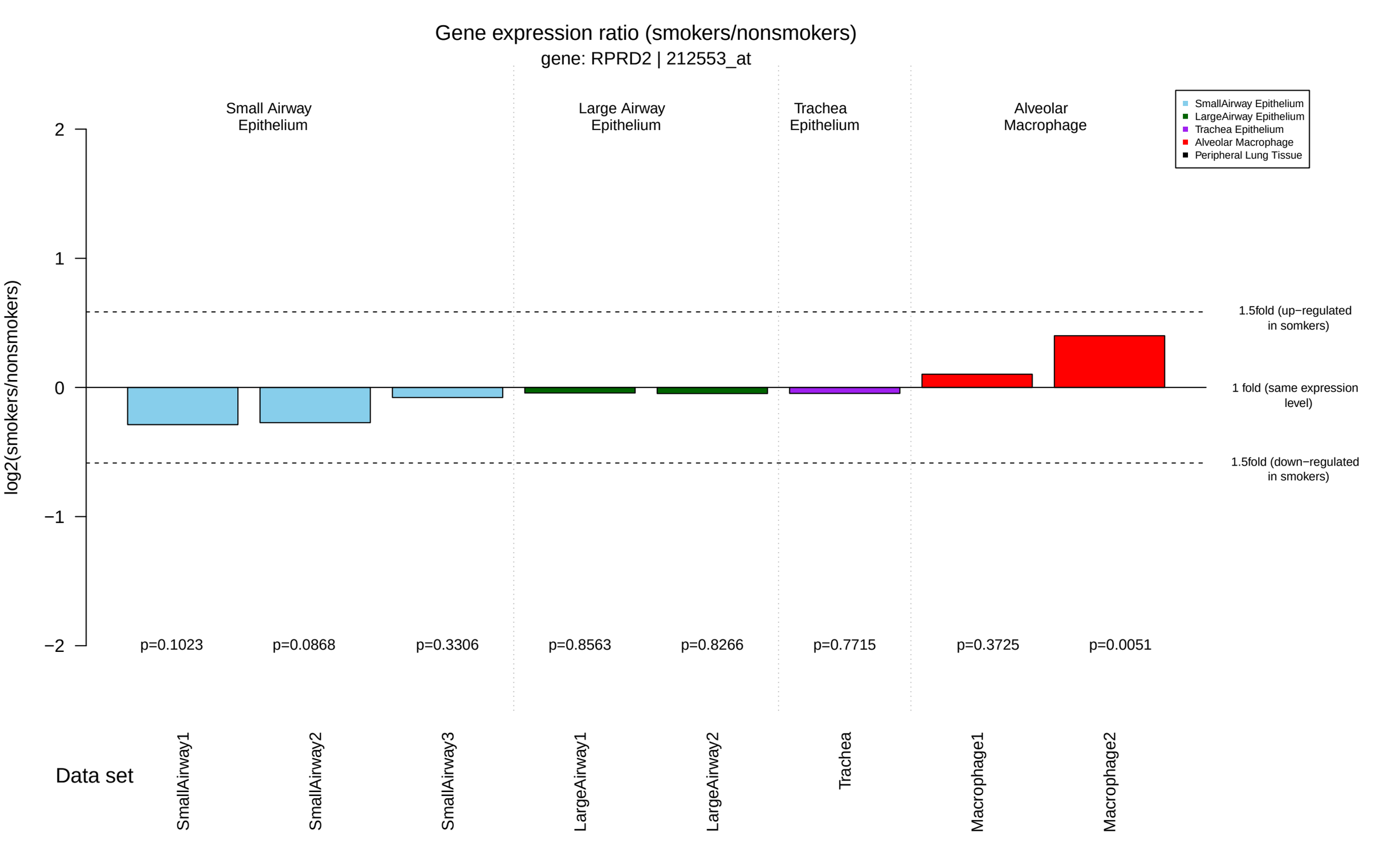
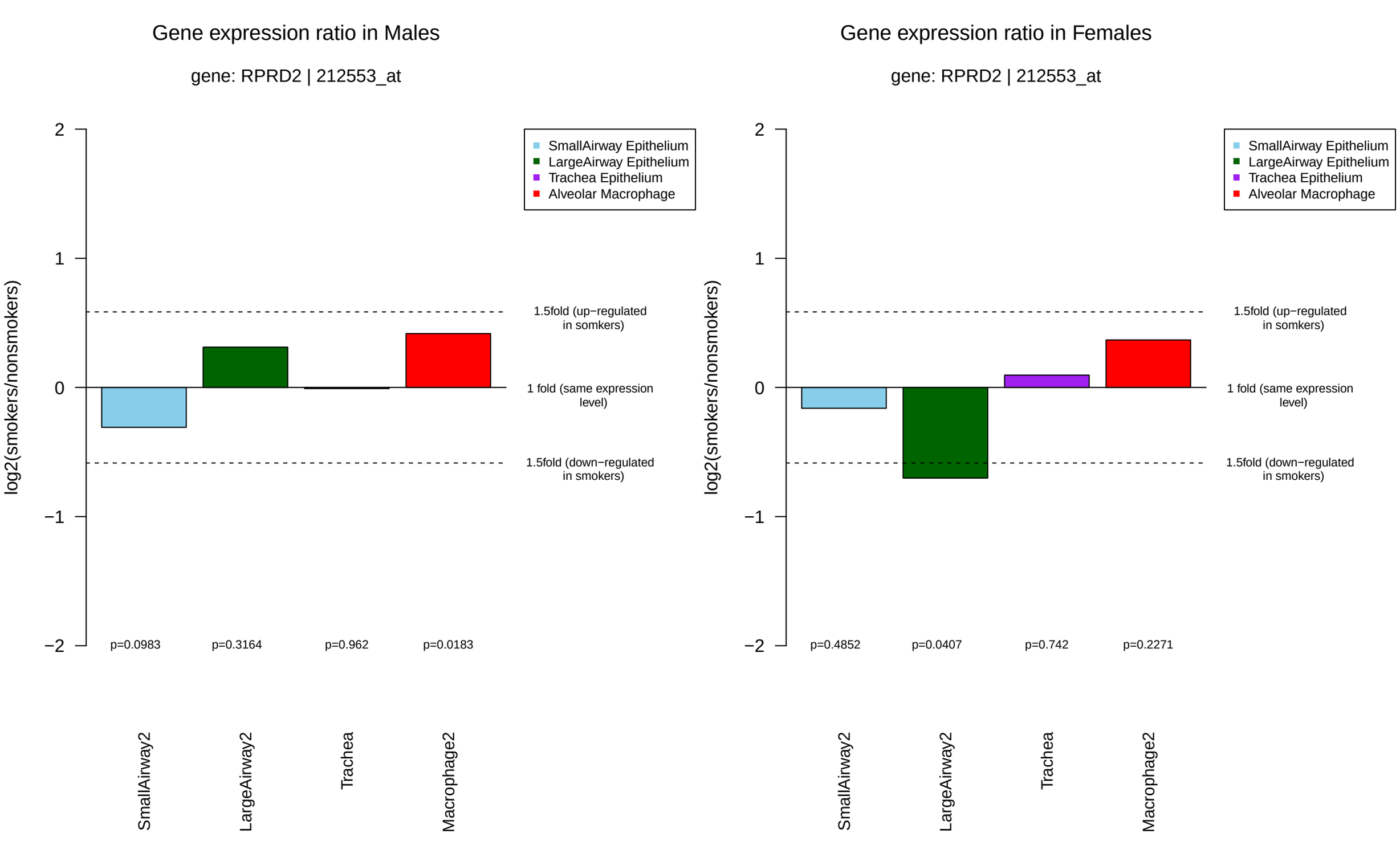
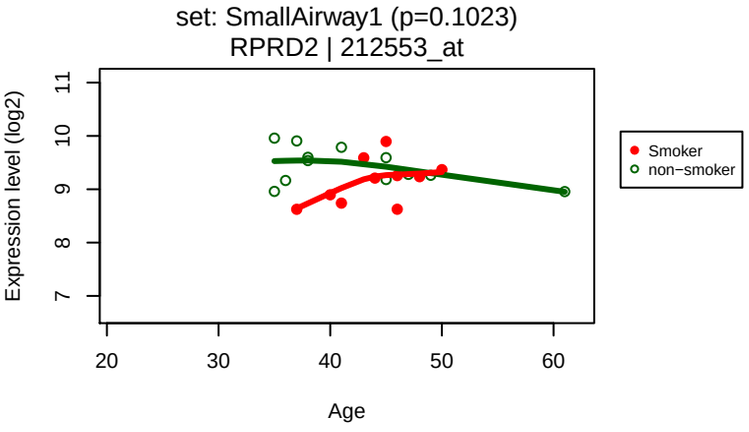
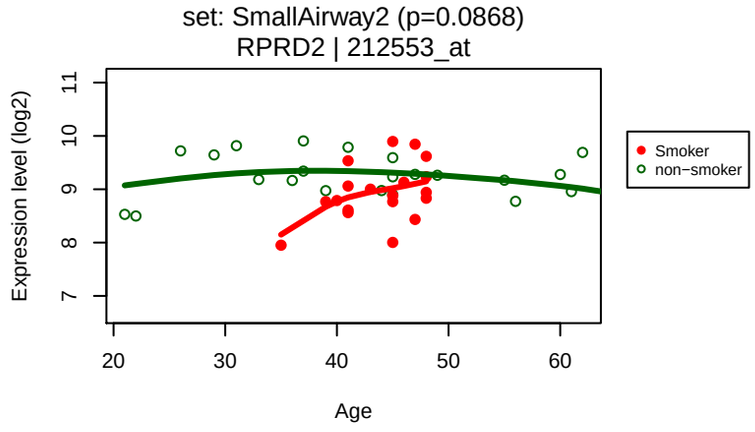
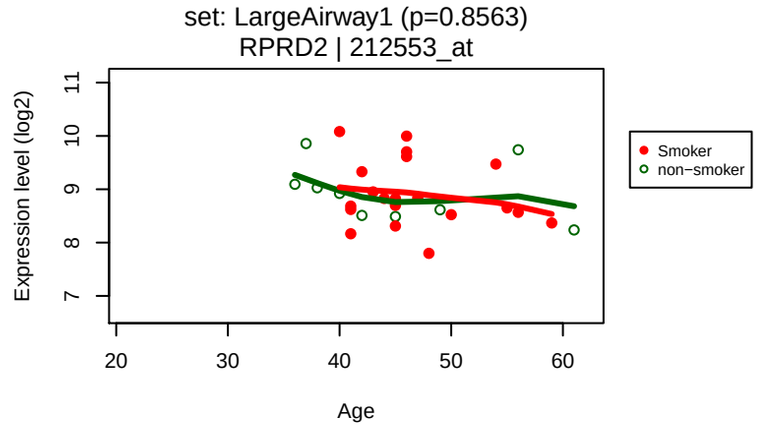

Gene expression of smokers and nonsmokers
gene: RPRD2 | 212553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1023
p=0.0868
p=0.3306
p=0.8563
p=0.8266
p=0.7715
p=0.3725
p=0.0051
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

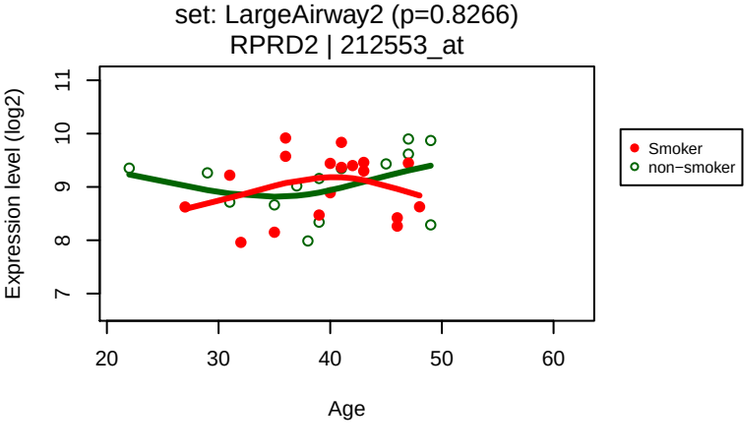

Gene expression ratio (smokers/nonsmokers)
gene: RPRD2 | 212553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1023
p=0.0868
p=0.3306
p=0.8563
p=0.8266
p=0.7715
p=0.3725
p=0.0051
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

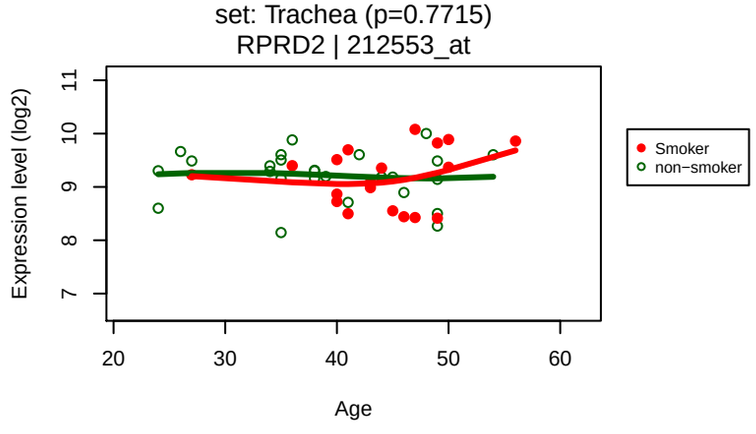

Gene expression ratio in Males
gene: RPRD2 | 212553_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0983
p=0.3164
p=0.962
p=0.0183
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPRD2 | 212553_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4852
p=0.0407
p=0.742
p=0.2271
SmallAirway2
LargeAirway2
Trachea
Macrophage2

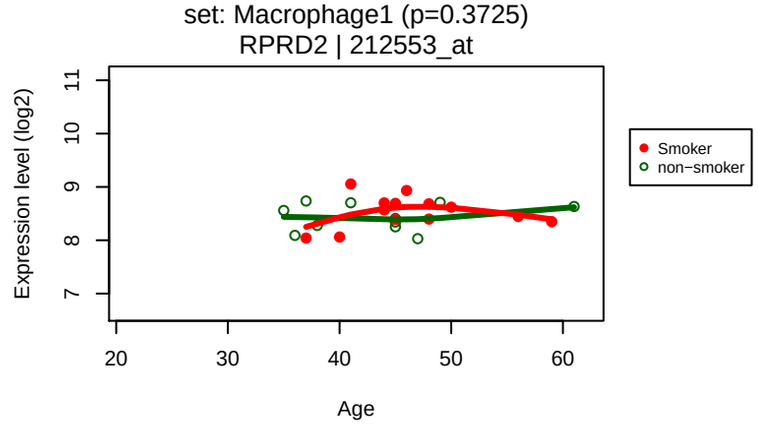

set: SmallAirway1 (p=0.1023)
RPRD2 | 212553_at
Expression level (log2)
Age
Smoker
non−smoker

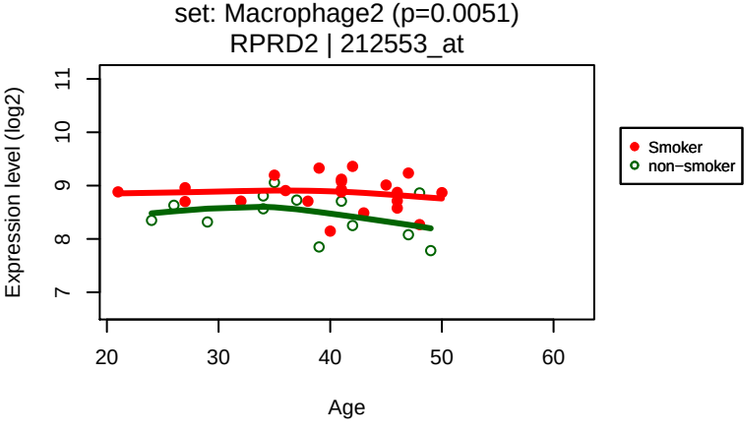

set: SmallAirway2 (p=0.0868)
RPRD2 | 212553_at
Expression level (log2)
Age
Smoker
non−smoker

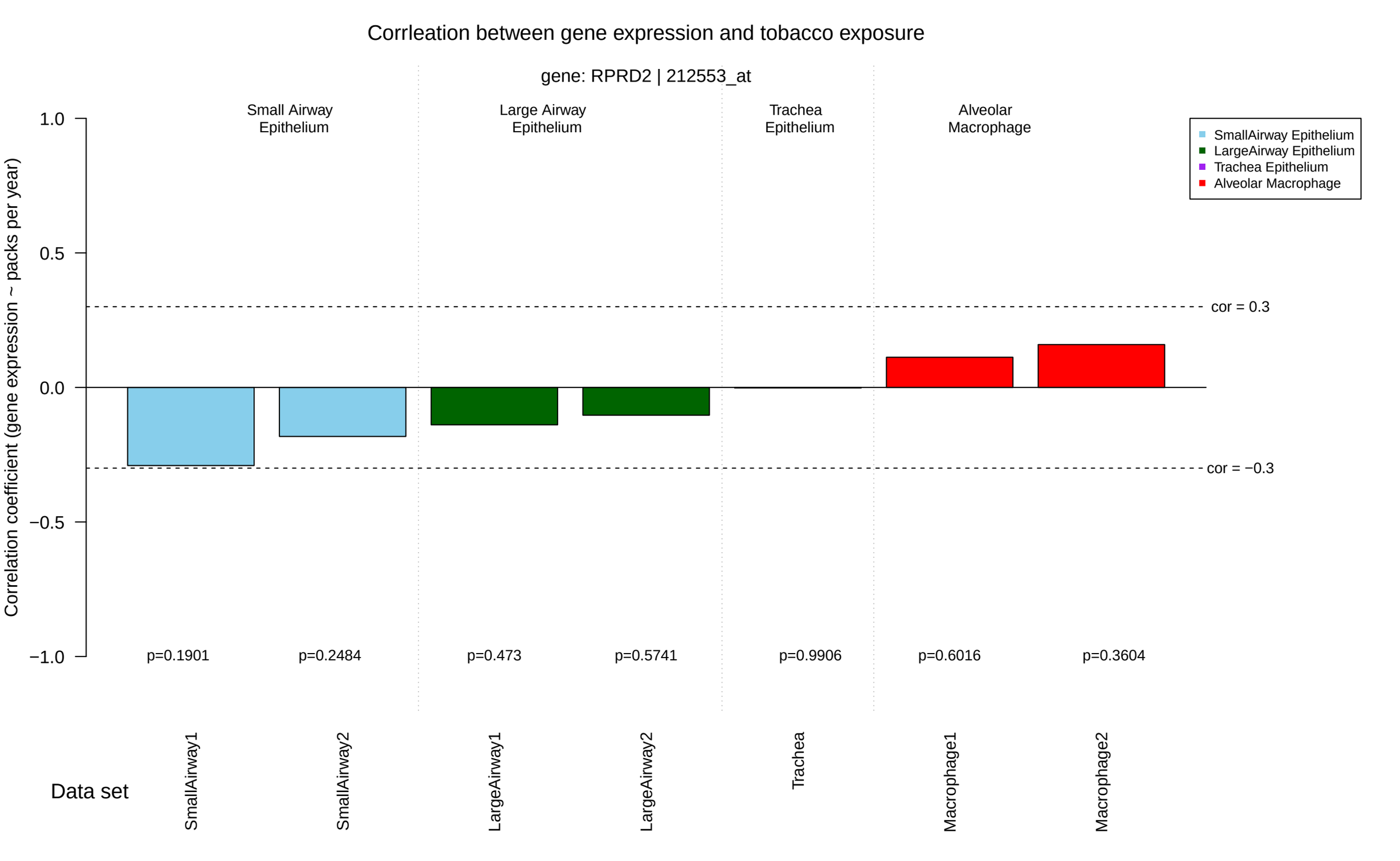

set: LargeAirway1 (p=0.8563)
RPRD2 | 212553_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8266)
RPRD2 | 212553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7715)
RPRD2 | 212553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3725)
RPRD2 | 212553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0051)
RPRD2 | 212553_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPRD2 | 212553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1901
p=0.2484
p=0.473
p=0.5741
p=0.9906
p=0.6016
p=0.3604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2